# BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

# BIOLOGY

Past Paper 2023 • Paper Code: 6461 / 6466

Retyped faithfully from the original board paper — same questions, no changes.

## GROUP – I

### Biology (Objective) Group - I

**Time Allowed:** 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6461

Note: Q.No.1. You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. The control of pests by some living organism is called:

(A) Pest control
(B) Living control
(C) Biological control
(D) Organismic control

Q2. The human bone cells contain the amount of water is:

(A) 5 %
(B) 10 %
(C) 15 %
(D) 20 %

Q3. Which of the enzyme has pH = 9?

(A) Pancreatic lipase
(B) Pancreatic amylase
(C) Chymotrypsim
(D) Arginase

Q4. The haploid number of chromosomes in drosophila melanogaster is:

(A) 2
(B) 4
(C) 8
(D) 16

Q5. The botanical name of tomato is:

(A) Solanum-nigrum
(B) Solanum-tuberosum
(C) Solanum-esculentum
(D) Solanum-melongena

Q6. Antonie Van Leeuwen Hock was the first to report the microbes:

(A) Bacteria and protozoa
(B) Bacteria and virus
(C) Bacteria and algae
(D) Bacteria and fungi

Q7. The animal live in the gut of termites is:

(A) Ciliate
(B) Trypanosome
(C) Choanoflagellates
(D) Trichonymphas

Q8. The multinucleated fungus group is:

(A) Zygomycota
(B) Ascomycota
(C) Basidiomycota
(D) Deuteromycota

Q9. Prosopis-glandulosa belongs to family:

(A) Rosaceae
(B) Mimosaceae
(C) Fabaceae
(D) Poaceae

Q10. The red blood cells of mammals are:

(A) Nucleated
(B) Many nucleated
(C) Non-nucleated
(D) Nucleated first then non-nucleated

Q11. The organ madreporite is found in:

(A) Coelenterates
(B) Annelids
(C) Molluscus
(D) Echinoderms

Q12. In cyclic phosphorylation electrons from primary acceptor of photosystem-I flows back to:

(A) Pq
(B) Pc
(C) Cytochrome complex
(D) NADP+

Q13. In prokaryotes the chlorophyll is present in:

(A) Photosynthetic membranes
(B) Stroma of chloroplast
(C) Thylakoid membranes
(D) Granum of chloroplast

Q14. The examples of parasitic plant is:

(A) Lichen
(B) Dodder
(C) Drosera
(D) Neotia

Q15. How much air, lungs can held when they are fully inflated:

(A) 3.5 litre
(B) 1.5 litre
(C) 5 litre
(D) 4 litre

Q16. The hydrostatic pressure in xylem is increased when root pressure:

(A) Increased
(B) Decreased
(C) Remain static
(D) Increased first then decreased

Q17. Antiserum is a serum containing:

(A) Antigens
(B) Antibodies
(C) Platelets
(D) Lymphocytes

## Biology (Subjective) Group - I

**Time Allowed:** 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 1115-XI123-38000

Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT parts.

(i) Lipids has double amount of energy as compared to same amount of carbohydrates. Why?

(ii) At pH2 pepsin works while arginase does not work. Why?

(iii) What are reversible inhibitors?

(iv) Differentiate between prosthetic group and coenzyme.

(v) What is candidosis?

(vi) How genetic recombination occurs in imperfect fungi?

(vii) Differentiate between ostia and osculum.

(viii) Define polymorphism.

(ix) What are beneficial insects?

(x) What is syrinx and where it is situated?

(xi) How entry of $CO_2$ into leaves is controlled?

(xii) Calvin cycle is called C3 pathway. Justify it.

Q3. (Marks 16) Write short answers to any EIGHT parts.

(i) What do you know about biome?

(ii) Give the two advantages of tissue culture techniques.

(iii) Differentiate between primary cell wall and secondary cell wall.

(iv) Suggest any two functions of Golgi complex.

(v) What do you know about kelps?

(vi) How does locomotion take place in apicomplexans?

(vii) What do you know about choanoflagellates?

(viii) Give the special features of giant amoeba.

(ix) Differentiate between protonema and paraphyses?

(x) Explain the term double fertilization.

(xi) Differentiate between symplast pathway and apoplast pathway.

(xii) Suggest the location and function of coronary artery.

Q4. (Marks 12) Write short answers to any SIX parts.

(i) What are pocks?

(ii) Define plasmids. How these are important?

(iii) What is saliva? Give its ingredients.

(iv) Compare saprophytic and parasitic mode of nutrition.

(v) What are omnivores? Give example.

(vi) Differentiate between epiglottis and glottis.

(vii) What are spiracles? Give their function.

(viii) Write two factors which affect transport of oxygen in blood.

(ix) How does respiration occur in earthworm?

> Note: SECTION - II

> Note: Attempt any THREE questions. Each question carries 08 marks.

Q5.

(a) [04] How can you solve biological problem with help of biological method?

(b) [04] Write a note on blood plasma.

Q6.

(a) [04] Write a note on importance of water.

(b) [04] Discuss economic losses due to fungi.

Q7.

(a) [04] Illustrate various methods to control bacteria in home, industry as well as in medical fields.

(b) [04] Write significance of alternation of generation in plants.

Q8.

(a) [04] What are small pox and herpes simplex diseases?

(b) [04] Draw the sketch of electron transport chain and chemiosmosis, coupling ETC and formation of ATP by chemiosmosis.

Q9.

(a) [04] What are plastids? Describe their different types.

(b) [04] Describe digestion of food in planaria.

## GROUP – II

### Biology (Objective) Group - II

**Time Allowed:** 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6466

Note: Q.No.1. You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. The oyster mushroom is considered as:

(A) Parasitic fungi
(B) Saprotrophic fungi
(C) Carnivorous fungi
(D) Omnivorous fungi

Q2. The type of algae takes part in coral reef formation is:

(A) Brown
(B) Green
(C) Golden
(D) Red

Q3. Which type of respiration is found in bacterium E.Coli?:

(A) Aerobic
(B) Anaerobic
(C) Micro aerophilic
(D) Facultative anaerobic

Q4. The mysterious brain infection is caused by:

(A) Virion
(B) Prion
(C) Bacteria
(D) Fungi

Q5. The protein present in microtubules is:

(A) Actin
(B) Myosin
(C) Tubulin
(D) Tropomysin

Q6. The vitamins are essential raw material for the synthesis of:

(A) Activator
(B) Co-factor
(C) Co-enzyme
(D) Prosthetic group

Q7. Monosaccharides which rare in nature and occur in some bacteria are:

(A) Troises
(B) Tetroses
(C) Pentoses
(D) Hexoses

Q8. The stomach can digest the:

(A) Proteins
(B) Fats
(C) Starch
(D) Cellulose

Q9. Which is agranulocyte?

(A) Basophilis
(B) Eosinophilis
(C) Monocyte
(D) Neutrophilis

Q10. The volume of blood in a body weight of 72 kg is:

(A) 5 litres
(B) 6 litres
(C) 7 litres
(D) 8 litres

Q11. About 70% of CO2 is carried as:

(A) Carbonates
(B) Bicarbonate
(C) Carbonic acid
(D) Acetic acid

Q12. Pepsinogen is secreted by:

(A) Mucous cells
(B) Parietal cells
(C) Zymogen cells
(D) Oxyntic cells

Q13. The first action spectrum was obtained by:

(A) Van Neil
(B) Engelmann
(C) Melvin Calvin
(D) Van Mohl

Q14. Chlorophyll-b is found along with chlorophyll-a in:

(A) Bacteria
(B) Cyanobacteria
(C) Bryophytes
(D) Embryophytes

Q15. The length of giant squid in meters is equal to:

(A) 5 (B) 10

(C) 15 (D) 20

Q16. The exclusively marine phylum is:

(A) Protozoa (B) Porifera

(C) Coelenterata (D) Echinodermata

Q17. The scientific name of egg plant is:

(A) Capsicum annum (B) Solanum tuberosum

(C) Solanum melangena (D) Solanum nigreem

## Biology (Subjective) Group - II

**Time Allowed:** 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 1116-XI123-2000

Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT parts.

(i) Differentiate between amylose and amylopectin.

(ii) Do you think that koshland change the idea of enzyme action?

(iii) Write any two properties of enzymes.

(iv) How heat influence the rate of enzyme action?

(v) What are two main ecological importances of Lichen's?

(vi) Differentiate parasitic fungi from saprophytic fungi.

(vii) Differentiate protostomes from deuterostomes.

(viii) Why arthropods have reached the peak of invertebrate evolution?

(ix) How swim bladder help the fish to survive in aquatic environment?

(x) Write any four characters of birds.

(xi) Define compensation point.

(xii) How is daily rhythmic activity of stomata governed?

Q3. (Marks 16) Write short answers to any EIGHT parts.

(i) Name and define the method used to protect walnut tree from pest.

(ii) The environmental pollution is a national problem in Pakistan. Why?

(iii) Cell membrane offers a barrier between cell content and environment. Justify it.

(iv) Golgi complex is concerned with cell secretions. Why?

(v) Why fungus like protists are not fungi?

(vi) Water mold played infamous role in human history. Justify it.

(vii) Write two uses of chlorella.

(viii) What are red tides?

(ix) Define protonema.

(x) What is the role of meristematic tissue in hornworts?

(xi) Differentiate between source and sink.

(xii) The heart of fish is single circuit. Why?

Q4. (Marks 12) Write short answers to any SIX parts.

(i) What are the symptoms of AIDS?

(ii) Differentiate between prions and virions.

(iii) What is hunger pang? Also write its causes.

(iv) How gastric juice production is regulated?

(v) Differentiate between oxyntic cells and zymogen cells.

(vi) What is diving reflex and how it is activated?

(vii) Suggest the various characteristics of respiratory surface.

(viii) What is respiratory distress syndrome?

(ix) How does respiration take place in Earthworm?

> Note: SECTION - II

> Note: Attempt any THREE questions. Each question carries 08 marks.

Q5.

(a) [04] Discuss biological method.

(b) [04] Give comparison between closed and open circulatory system.

Q6.

(a) [04] What are polysaccharides? Discuss in detail.

(b) [04] Give economic losses due to fungi.

Q7.

(a) [04] Discuss the structure of bacterial cell wall. Also give comparison between Gram positive and negative bacteria.

(b) [04] Discuss the life cycle of Moss.

Q8.

(a) [04] Explain five kingdom system of classification in detail. Discuss its modification as well.

(b) [04] Discuss various photosynthetic pigments in detail.

Q9.

(a) [04] What role plays cytoskeleton in the cell?

(b) [04] Discuss parasitic nutrition and its various types.